ASCP Plan Failure

Memory Based Snapshot Workers (MSCNSP) failed during the overnight refresh, blocking master scheduling and producing wrong ATP for new sales orders.

PRIORITY	MODULE	STATUS	REQUEST
P2 / High	ASCP	ERROR	8847123

Customer-facing problem

Module	Oracle Advanced Supply Chain Planning (ASCP)
Reported by	Master Scheduler, NA Manufacturing
Reported at	2026-05-13 07:48 local
Severity	P2 (High). Production meeting at 10:00 is at risk.
Business impact	Master scheduling dashboards on stale data. ATP wrong for at least three customer orders. Order Management already escalating.

CUSTOMER SUMMARY (VERBATIM)

> *The overnight ASCP plan did not refresh. Our master scheduling dashboards are showing yesterday's data, ATP for new sales orders is wrong, and the 03:42 batch failed with a database error. We need this resolved before the 10:00 production meeting. Order Management is already escalating three customer orders with bad promise dates.*

The failed concurrent program

Program (short name)	MSCNSP
Program (long name)	Memory Based Snapshot Workers
Request ID	8847123
Phase / Status	COMPLETED / ERROR (Phase=C, Status=E)
Submitted by	APPS
Submitted at	2026-05-13 03:42:18
Actual start	2026-05-13 03:42:31
Actual completion	2026-05-13 03:58:04
Plan name	ASCP_NA_MFG_PLAN
Plan ID	401
Snapshot mode	MEMORY_BASED
Completion text	Concurrent Manager encountered an error during snapshot phase.

Request log excerpt

Relevant lines from the request log (FND_CONCURRENT_REQUESTS.LOGFILE_NAME). The error is an ORA-01555 raised inside a parallel query worker.

```
03:42:31 INFO  MSCNSP-1010: Memory Based Snapshot starting for plan ASCP_NA_MFG_PLAN
03:42:33 INFO  MSCNSP-2003: Spawning 8 parallel workers
03:51:18 ERROR ORA-12801: error signaled in parallel query server P008
03:51:18 ERROR ORA-01555: snapshot too old: rollback segment number 14
                with name "_SYSSMU14_3092830574$" too small
03:51:19 ERROR MSCNSP-9001: Worker P008 failed reading MSC_SYSTEM_ITEMS
03:51:19 ERROR MSCNSP-9100: Snapshot phase aborted. No partial data committed.
03:58:04 INFO  Program completed with error.
```

MSC_PLAN_RUNS for plan 401

The plan has not had a successful run since 2026-05-11. The last two attempts both failed in the snapshot phase.

PLAN_RUN_ID	PLAN_ID	START_DATE	END_DATE	STATUS
2147	401	2026-05-11 03:42:18	2026-05-11 04:38:51	COMPLETE
2148	401	2026-05-12 03:42:14	2026-05-12 03:55:09	ERROR
2149	401	2026-05-13 03:42:31	2026-05-13 03:58:04	ERROR

MSC_EXCEPTION_DETAILS raised by the failed runs

EXCEPTION_TYPE	COUNT	FIRST_SEEN
PLAN_SNAPSHOT_INCOMPLETE	1	2026-05-12 03:55:09
DEMAND_NOT_REFRESHED	1	2026-05-12 03:55:09
ATP_BASIS_STALE	1	2026-05-13 03:58:04

MSC_DEMANDS staleness

The MSC_DEMANDS table is 47,832 rows. Last refresh was 2026-05-11 03:55 (the last successful run). Any ATP query is currently reading data that is 50 hours old.

Standard Collection runs on the source instance

The Standard Collection job that populates MSC_SYSTEM_ITEMS, MSC_BOMS, and MSC_RESOURCES before the snapshot runs.

REQUEST_ID	SUBMITTED	STATUS
8841902	2026-05-11 23:42	COMPLETE

REQUEST_ID	SUBMITTED	STATUS
8845671	2026-05-12 23:42	ERROR
8847002	2026-05-13 03:25	RUNNING

The 23:42 collection on 2026-05-12 failed (which is why the 03:42 snapshot also failed). Someone resubmitted at 03:25 today, and that resubmission is still running. It started 17 minutes before the snapshot, then both processes were touching MSC_SYSTEM_ITEMS at the same time.

Likely root cause

Two issues stacked on top of each other.

1. **UNDO_RETENTION is too low.** The database undo retention is set to 900 seconds (15 minutes). The Memory Based Snapshot in parallel mode reads MSC_SYSTEM_ITEMS, MSC_BOMS, and MSC_RESOURCES with a read-consistent query that takes 8 to 10 minutes. When the concurrent Standard Collection on instance 2 updated MSC_SYSTEM_ITEMS during that read window, the snapshot lost its rollback view and worker P008 raised ORA-01555.
2. **Standard Collection 23:42 schedule is too close to Snapshot 03:42.** When the 23:42 collection runs long (because of overnight Item Open Interface batches feeding MSC_SYSTEM_ITEMS), the operator pattern is to resubmit in the morning. That resubmission collides with the 03:42 snapshot window. Once the resubmission was triggered at 03:25, the collision was guaranteed.

PRIMARY VS PROXIMATE

The primary cause is the schedule collision. The undo retention is the proximate cause of the error, but raising it without fixing the schedule will only push the problem out by a few minutes.

Recommended fix

1. **Resubmit the failed snapshot** once the in-progress collection completes. ASCP Plan Workbench, Plan ASCP_NA_MFG_PLAN, then Plan, Launch Plan, with Snapshot phase enabled. Or resubmit request 8847123 from System Administrator, Concurrent, Requests.
2. **Increase UNDO_RETENTION** on the destination database to 3600 seconds (one hour) and confirm the UNDO tablespace has AUTOEXTEND on. SQL: ALTER SYSTEM SET UNDO_RETENTION=3600 SCOPE=BOTH; then verify with SELECT TUNED_UNDORETENTION FROM V$UNDOSTAT WHERE ROWNUM=1.
3. **Reschedule Standard Collection** to complete at least 60 minutes before the Snapshot runs. Move the nightly Standard Collection from 23:42 to 22:00. Confirm there is no other process that updates MSC_SYSTEM_ITEMS during the 03:00 to 04:00 window.
4. **Stop manual resubmissions during the snapshot window.** Add an operational rule: if Standard Collection fails, alert the DBA and skip that day's snapshot rather than resubmit at 03:25.
5. **Add monitoring.** Configure an alert on MSC_PLAN_RUNS so any plan run with STATUS=ERROR pages the on-call DBA. Finding out at 07:48 when master schedulers open their dashboards is too late.
6. **Verify after the run.** Confirm MSC_PLAN_RUNS shows STATUS=COMPLETE, MSC_DEMANDS row count matches expected (about 48,000), and MSC_EXCEPTION_DETAILS no longer shows PLAN_SNAPSHOT_INCOMPLETE for plan 401.

Related Oracle MOS docs

- Doc ID 1326260.1: How to Diagnose ASCP Memory Based Snapshot Errors
- Doc ID 432384.1: ORA-01555 Snapshot Too Old in ASCP Plan Run
- Doc ID 1357887.1: ASCP Plan Run Performance Tuning
- Doc ID 269021.1: Troubleshooting ASCP Collections
- Doc ID 1083387.1: Configuring UNDO_RETENTION for Long-Running Reports

How to use this in the demo

Use this scenario for Scene 1 (reactive flow). It is specific enough to feel real and short enough to fit in four minutes.

Recommended live flow

1. Pre-create a JIRA ticket in the SUP project titled *ASCP overnight plan run failed. Master scheduling on stale data.* Paste the customer summary as the description.
2. On stage: open the ticket from the JIRA page in this app. Show the customer summary in the drawer.
3. Click **Troubleshoot in EBS**. While Claude is working, explain what is happening: the model is reading the ticket text, classifying it as a planning issue, and choosing the right Agentic Apps from the catalog.
4. When the report renders, walk the audience through the four sections. The Likely root cause section is the moneymaker: naming UNDO_RETENTION, MSC_SYSTEM_ITEMS, and the schedule collision should mirror the analysis in this document.
5. Click **Post Report as JIRA Comment**. Switch to JIRA web. Show the formatted comment with headings, bold table names, and a numbered fix list.

AUDIENCE QUESTION TO PLANT

If someone asks how the AI knew about UNDO_RETENTION, the honest answer is that it has the EBS knowledge baked in and the SQL evidence in front of it. The combination is what produces credible, specific analysis. Take that question. It is the right one to ask.